Reddit Sentiment Analysis

Project Report

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BS Artificial Intelligence

Date: October 2025

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0.1 Introduction

The project **"Reddit Sentiment Analysis"** using SageMaker and Streamlit focuses on building an automated sentiment classification system to analyze Reddit posts and classify them into different categories (flairs). The project combines the power of machine learning models, data preprocessing, and cloud-based model training using Amazon SageMaker.

The system enables efficient prediction of sentiment and content type from Reddit posts by leveraging TF-IDF-based feature extraction and traditional machine learning models such as Logistic Regression, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Finally, the model is deployed using Streamlit with ngrok for easy web access.

0.2 Objectives

The main objectives of this project are:

- To collect Reddit data through web scraping using Selenium and BeautifulSoup
- To preprocess and vectorize text data using TF-IDF for machine learning
- To train multiple classification models and evaluate their performance using comprehensive metrics
- To deploy the sentiment classification model using Streamlit with public access via ngrok
- To integrate cloud-based training and prediction using AWS SageMaker for scalable processing
- To compare model performance between local training and cloud-based approaches

0.3 Tools and Technologies

- **Programming Language:** Python
- **Machine Learning Libraries:** scikit-learn, pandas, numpy, joblib
- **Web Scraping:** BeautifulSoup, Selenium WebDriver
- **Cloud Platform:** Amazon SageMaker, AWS S3, AWS IAM

- **Deployment Tools:** Streamlit, ngrok, Flask
- **Data Visualization:** matplotlib, seaborn
- **Version Control:** Git, GitHub

0.4 Dataset Description

The dataset was obtained using a custom Reddit web scraper built with Selenium and BeautifulSoup that handles dynamic content loading and pagination. The comprehensive dataset includes:

0.4.1 Data Collection Process

- Automated navigation through Reddit pages using Selenium WebDriver
- Parsing HTML content with BeautifulSoup for structured data extraction
- Handling of dynamic content loading with appropriate wait times
- Implementation of error handling and rate limiting to avoid IP blocking

0.4.2 Data Features

- **Text Content:** Post titles and bodies containing the main discussion content
- **Engagement Metrics:** Upvotes, downvotes, and comment counts measuring user interaction
- **Metadata:** Timestamps, author information, and post URLs
- **Target Variable:** Flair categories (Discussion, Project, Research, etc.) for classification
- **Subreddit Information:** Source community context for each post

The final curated dataset was stored as a CSV file containing cleaned textual data and engineered features, ready for machine learning processing.

0.5 Methodology

The project follows a systematic machine learning pipeline with the following key stages:

0.5.1 Data Preprocessing Pipeline

- **Text Cleaning:** Removal of special characters, URLs, and non-ASCII characters
- **Tokenization:** Splitting text into individual words and phrases
- **Stop Word Removal:** Elimination of common words that don't contribute to meaning
- **Lemmatization:** Reducing words to their base or dictionary form
- **TF-IDF Vectorization:** Application of Term Frequency-Inverse Document Frequency on post titles with maximum features set to 5000
- **Label Encoding:** Conversion of categorical flair labels into numerical representations using LabelEncoder
- **Feature Integration:** Combination of text-based TF-IDF features with additional numerical attributes

0.5.2 Data Visualization

Several advanced visualization techniques were employed to gain insights into the dataset structure and feature relationships. The generated plots are stored as PNG files in the project's visualization directory.

- **Correlation Heatmap:** A heatmap visualization was created to analyze feature correlations within the dataset. This plot uses a color-gradient scheme where strong positive correlations appear in dark red, strong negative correlations in dark blue, and neutral correlations in light colors. The mathematical basis is Pearson's correlation coefficient:

  $$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

  The resulting visualization `heatmap.png` revealed multicollinearity patterns and guided feature selection decisions for the machine learning pipeline.

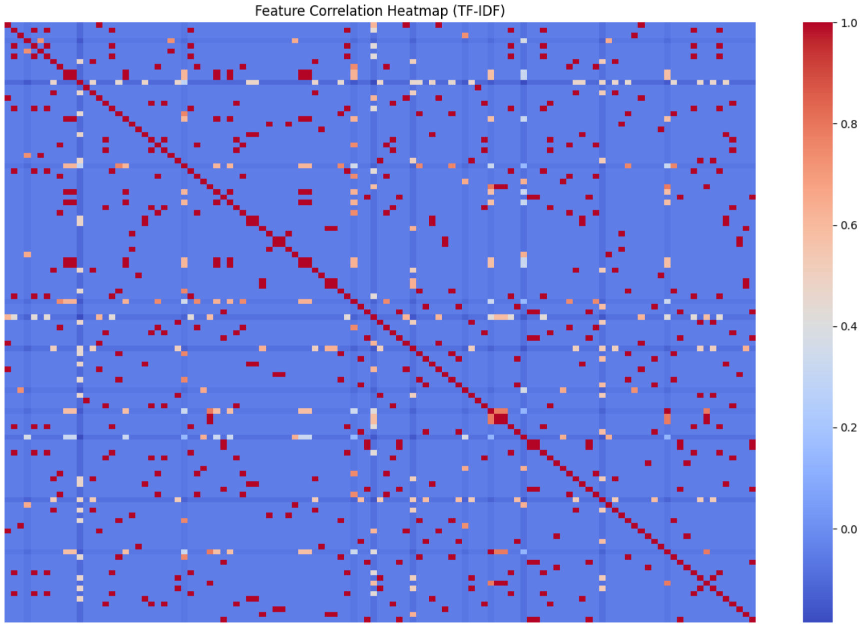

Figure 1: Heatmap of all Numerical features

- **PCA (Principal Component Analysis):** PCA was applied to reduce the high-dimensional TF-IDF feature space while maximizing variance retention. The transformation involves eigen decomposition of the covariance matrix:

$$\Sigma = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

  The 2D projection 2 visualize the data distribution in reduced dimensional space, showing cluster separation between different flair categories and revealing the underlying data structure.

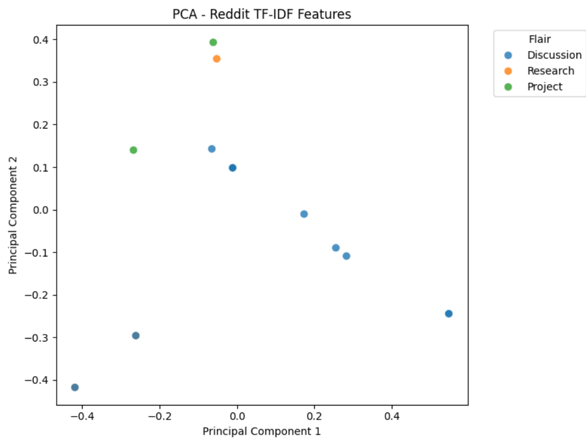

Figure 2: 2-D Principal Component Analysis

- **t-SNE (t-Distributed Stochastic Neighbor Embedding):** t-SNE was utilized for nonlinear dimensionality reduction to preserve local data structures. The algorithm minimizes the Kullback-Leibler divergence between probability distributions:

$$KL(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

where p_{ij} and q_{ij} represent similarity probabilities in high and low dimensions respectively. The visualization 3 effectively captured local clustering patterns and revealed nuanced relationships between data points that were not apparent in linear projections.

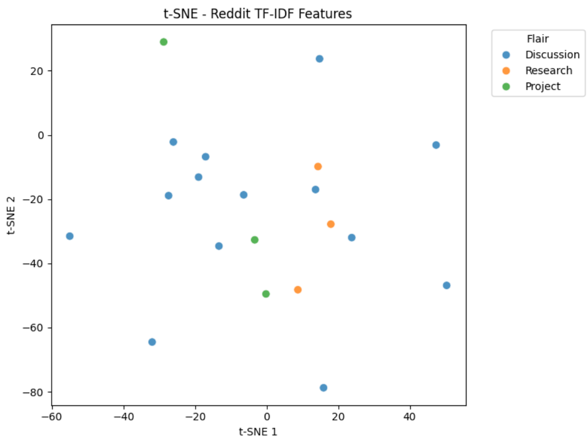

Figure 3: TSNE

0.5.3 Machine Learning Models

Three supervised machine learning models were trained and evaluated:

Logistic Regression

- Implementation with L2 regularization to prevent overfitting
- Configuration with balanced class weights to handle imbalanced data
- Optimization using cross-validation and hyperparameter tuning

Support Vector Machine (SVM)

- Linear kernel implementation for efficient text classification

- Parameter optimization for regularization and kernel parameters
- Handling of high-dimensional feature spaces from TF-IDF vectors

K-Nearest Neighbors (KNN)

- Distance-based classification with Euclidean distance metric
- Optimization of k-value through cross-validation
- Feature scaling for improved distance calculation

0.5.4 AWS SageMaker Integration

The cloud integration component involved:

Data Management

- Secure upload of training and testing datasets to AWS S3 bucket.
- Proper data partitioning and organization for efficient. access
- Implementation of data versioning and backup strategies.

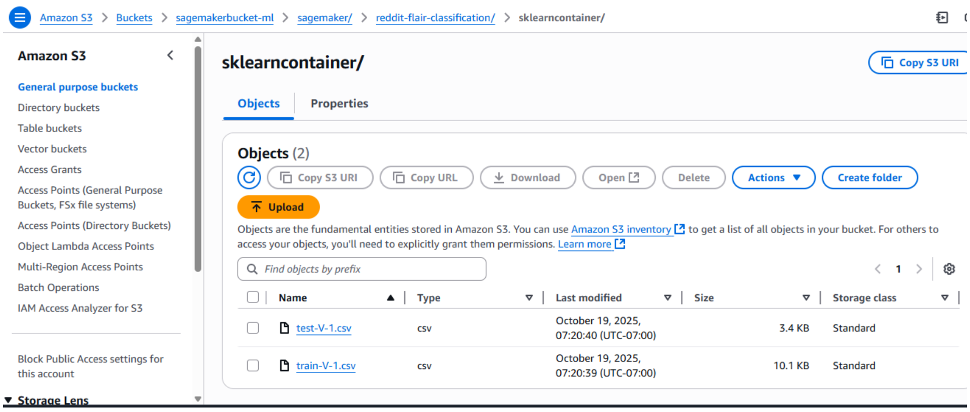

Figure 4: S3 Bucket for Data Storage

Model Training on SageMaker

- Configuration of SageMaker training jobs using scikit-learn containers

- Implementation of distributed training for large datasets
- Hyperparameter optimization using SageMaker's tuning capabilities
- Real-time monitoring of training progress through CloudWatch metrics

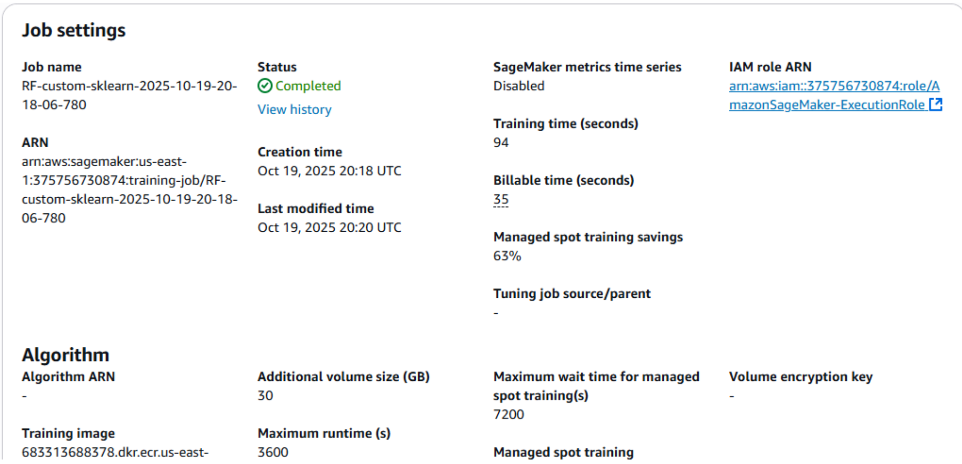

Figure 5: Model Training

Deployment and Inference

- Creation of SageMaker endpoints for model deployment
- Implementation of REST API for real-time predictions
- Configuration of auto-scaling for handling variable workloads
- Cost optimization through appropriate instance selection

0.5.5 Streamlit Deployment with Ngrok

The deployment phase included:

Web Application Development

- Interactive user interface design using Streamlit components
- Real-time input processing and prediction display
- User-friendly error handling and feedback mechanisms

Public Access Configuration

- Ngrok tunneling for secure public URL generation
- HTTPS configuration for secure data transmission
- Session management and request rate limiting

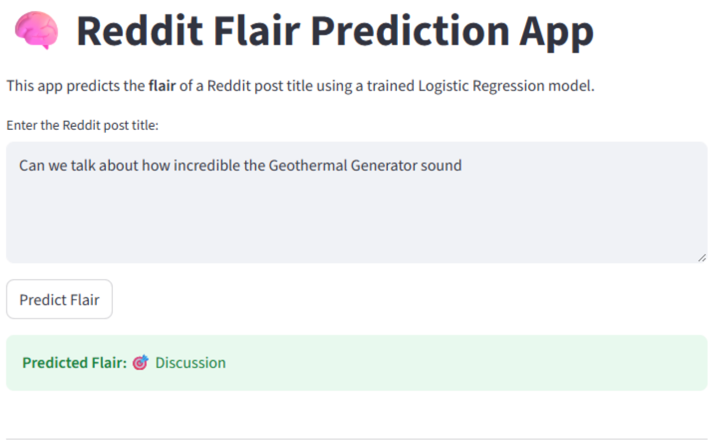

Figure 6: GUI

0.6 Evaluation and Results

The models were rigorously evaluated using multiple metrics including precision, recall, F1-score, and accuracy with stratified train-test splits.

0.6.1 Performance Metrics

K-Nearest Neighbors (KNN) Results

Class	Precision	Recall	F1-Score	Support
Discussion	0.60	1.00	0.75	3.0
Project	0.00	0.00	0.00	1.0
Research	0.00	0.00	0.00	1.0
Accuracy		**0.60**		5.0

Macro Avg	0.20	0.33	0.25	5.0
Weighted Avg	0.36	0.60	0.45	5.0

Logistic Regression Results

Class	Precision	Recall	F1-Score	Support
Discussion	0.60	1.00	0.75	3.0
Project	0.00	0.00	0.00	1.0
Research	0.00	0.00	0.00	1.0
Accuracy		**0.60**		5.0
Macro Avg	0.20	0.33	0.25	5.0
Weighted Avg	0.36	0.60	0.45	5.0

Support Vector Machine Results

Class	Precision	Recall	F1-Score	Support
Discussion	0.60	1.00	0.75	3.0
Project	0.00	0.00	0.00	1.0
Research	0.00	0.00	0.00	1.0
Accuracy		**0.60**		5.0
Macro Avg	0.20	0.33	0.25	5.0
Weighted Avg	0.36	0.60	0.45	5.0

0.6.2 Performance Analysis

- **Overall Accuracy:** All models achieved approximately 60% accuracy on the test dataset
- **Class-wise Performance:** Strong performance on "Discussion" class with perfect recall, but challenges with minority classes ("Project" and "Research")
- **Model Comparison:** Logistic Regression and SVM demonstrated similar performance characteristics, while KNN showed limitations with high-dimensional text data

- **Computational Efficiency:** Logistic Regression provided the best balance of performance and computational requirements

0.6.3 Limitations and Challenges

- **Data Imbalance:** Uneven distribution of flair categories affected minority class performance
- **Feature Sparsity:** High-dimensional TF-IDF features created challenges for distance-based algorithms
- **Sample Size:** Limited evaluation dataset size affected statistical significance of results
- **Real-time Performance:** Streamlit deployment faced latency challenges with model loading

0.7 Conclusion

This project successfully demonstrated an end-to-end sentiment analysis system for Reddit posts using traditional machine learning models enhanced with AWS SageMaker integration for scalable cloud-based model training. The Streamlit-based deployment with ngrok tunneling ensured a user-friendly interface for real-time prediction with public accessibility.

Key achievements include:

- Development of a robust web scraping pipeline for Reddit data collection
- Implementation of comprehensive text preprocessing and feature engineering
- Successful integration of local and cloud-based training approaches
- Deployment of a functional web application for real-time predictions
- Comparative analysis of multiple machine learning algorithms

0.8 Future Work

This project can be enhanced through several key improvements. Data collection would benefit from integrating the official Reddit API for reliable access and expanding to multiple subreddits for broader analysis. Model performance could be improved by implementing deep learning approaches like BERT and exploring ensemble methods. The deployment could be upgraded with permanent cloud hosting on AWS EC2 and user authentication features. Additional capabilities could include multilingual support for international communities and real-time sentiment tracking for trend analysis.